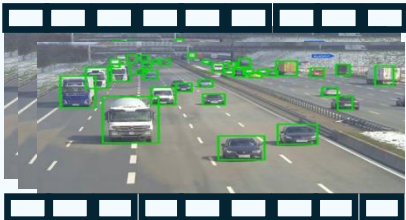


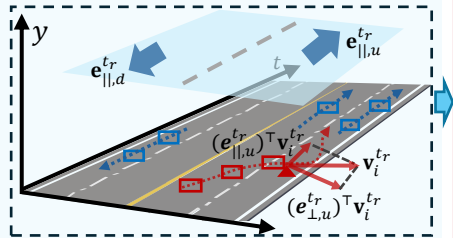
Vehicle Trajectory Tracking



Kinematics
Extraction

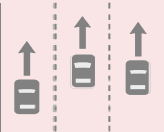
$$\{(t_r, \mathbf{b}_i^{tr})\}_{r=1}^{n_i} \rightarrow \{(t_r, \mathbf{v}_i^{tr})\}_{r=1}^{n_i}$$

Frenet Frame Formulation



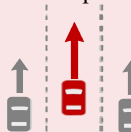
Kinematics-Guided Bayesian Inference

Normal-Motion



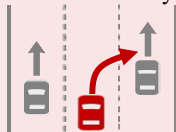
Accept

Overspeed



Reject

Lateral Anomaly

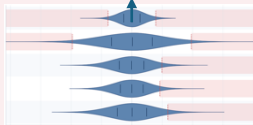


Reject

Hypothesis Test: $H_{0,i,d}^t: \mathbf{X}_{i,d}^t | \mathcal{H}_{i,d}^t, \mathcal{O}_{i,d}^t \sim \mathcal{N}(\hat{\mu}_{i,d}^t \mathbf{1}_L, \Sigma_{i,d}^t)$

Online Normal-Motion Estimation

Normal-Motion Distribution



Probability Boundary



Focused VLM Reasoning



$$[t_e^{pk} - \tau_e^-, t_e^{pk} + \tau_e^+]$$

Text Prompt:

Are there any traffic anomalies?



VLM

Answer:

Yes, a yellow car is swerving sharply.